```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import pandas as pd
```

```
putanja = '/content/drive/MyDrive/Colab Notebooks/news.csv'

news = pd.read_csv (putanja)
pd.set_option("display.max_colwidth", None)
```

Dataset news s primjerima članaka i njihovom klasifikacijom u jednu od 4 kategorije

```
news.head(20)
```

	text	category
0	The prime minister met with regional leaders to discuss energy policy and border security.	politics
1	Parliament approved a new education reform after a long debate among coalition members.	politics
2	The president called for national unity during a speech on foreign policy and defense.	politics
3	Lawmakers introduced a bill aimed at reducing taxes for small municipalities.	politics
4	The opposition party criticized the government over healthcare funding and public spending.	politics
5	A diplomatic meeting focused on trade relations, migration, and regional stability.	politics
6	The senate voted on judicial reforms and changes to electoral rules.	politics
7	Government officials announced a new housing program for young families.	politics
8	The mayor defended the city budget during a tense council session.	politics
9	Ministers discussed climate policy, public transport, and energy prices.	politics
10	The foreign minister visited neighboring countries to strengthen political cooperation.	politics
11	A parliamentary committee opened an investigation into public procurement procedures.	politics
12	The football team won the match after scoring two goals in the second half.	sports
13	The coach praised the players for their discipline and strong defense.	sports
14	A late penalty decided the derby between the top two clubs in the league.	sports
15	The tennis player reached the final after a dominant performance in straight sets.	sports
16	The basketball team improved its record with a win over a direct rival.	sports
17	Fans celebrated as the captain returned from injury and scored the winning goal.	sports
18	The striker signed a new contract after an impressive season with the club.	sports
19	The race ended dramatically when the leader lost control near the finish line.	sports

```
news.value_counts("category")
```

	count
category	
business	12
politics	12
sports	12
technology	12

dtype: int64

Koristit ćemo logističku regresiju jer ona daje težinske vrijednosti riječima

Koristimo običan BoW, kolika je točnost?

```
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, accuracy_score, confusion_matrix

# train/test split
X_train, X_test, y_train, y_test = train_test_split(
    news["text"],
    news["category"],
    test_size=0.25,
    random_state=42,
    stratify=news["category"]
)

# Bag of Words
vectorizer = CountVectorizer(stop_words="english")

X_train_bow = vectorizer.fit_transform(X_train)
X_test_bow = vectorizer.transform(X_test)

# model
model = LogisticRegression(max_iter=1000)
model.fit(X_train_bow, y_train)
```

```
# predictions
y_pred = model.predict(X_test_bow)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification report:\n")
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.5833333333333334

Classification report:

              precision    recall  f1-score   support

    business       0.50      0.67      0.57         3
    politics       1.00      0.33      0.50         3
      sports       0.50      0.33      0.40         3
  technology       0.60      1.00      0.75         3

    accuracy                           0.58        12
   macro avg       0.65      0.58      0.56        12
weighted avg       0.65      0.58      0.56        12
```

Ubacujemo ngrame, je li došlo do promjene točnosti?

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, accuracy_score

# Bag of Words + n-grams
vectorizer_ng = CountVectorizer(stop_words="english", ngram_range=(1,2))

X_train_ng = vectorizer_ng.fit_transform(X_train)
X_test_ng = vectorizer_ng.transform(X_test)

model_ng = LogisticRegression(max_iter=1000)
model_ng.fit(X_train_ng, y_train)

y_pred_ng = model_ng.predict(X_test_ng)

print("Accuracy with n-grams:", accuracy_score(y_test, y_pred_ng))
print("\nClassification report with n-grams:\n")
print(classification_report(y_test, y_pred_ng))
```

```
Accuracy with n-grams: 0.6666666666666666

Classification report with n-grams:

              precision    recall  f1-score   support

    business       0.50      0.67      0.57         3
    politics       1.00      0.33      0.50         3
      sports       0.50      0.67      0.57         3
  technology       1.00      1.00      1.00         3

    accuracy                           0.67        12
   macro avg       0.75      0.67      0.66        12
weighted avg       0.75      0.67      0.66        12
```

Dodatno pitanje za ponavljanje - koja je razlika izmedju precision i recall?

```
print("Number of features with simple BoW:", X_train_bow.shape[1])
print("Number of features with n-grams:", X_train_ng.shape[1])
```

```
Number of features with simple BoW: 243
Number of features with n-grams: 482
```

```
new_articles = [
    "The minister discussed tax reform and foreign policy during a parliamentary debate.",
    "The striker scored twice and the team won the match in the final minutes.",
    "The company released a software patch to fix a serious security issue.",
    "The bank reported stronger earnings and lower operating costs this quarter."
]

new_X = vectorizer_ng.transform(new_articles)
predictions = model_ng.predict(new_X)

for article, pred in zip(new_articles, predictions):
    print("TEXT:", article)
    print("PREDICTED CATEGORY:", pred)
    print("-" * 60)
```

```
TEXT: The minister discussed tax reform and foreign policy during a parliamentary debate.
PREDICTED CATEGORY: politics
------------------------------------------------------------
TEXT: The striker scored twice and the team won the match in the final minutes.
PREDICTED CATEGORY: sports
------------------------------------------------------------
TEXT: The company released a software patch to fix a serious security issue.
PREDICTED CATEGORY: technology
------------------------------------------------------------
TEXT: The bank reported stronger earnings and lower operating costs this quarter.
PREDICTED CATEGORY: business
------------------------------------------------------------
```

Dodatak - u grupama pokupajte utvrditi koji algoritam je bolji, logistička regresija ili MultinomialNB

Korištenjem BOW uz MultinomialNB

```
# --- LIBRARIES ---
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
```

```
from sklearn.metrics import classification_report, accuracy_score


# --- TRAIN / TEST SPLIT ---
X_train, X_test, y_train, y_test = train_test_split(
    news["text"],
    news["category"],
    test_size=0.25,
    random_state=42,
    stratify=news["category"]
)

# --- BAG OF WORDS ---
vectorizer_bow = CountVectorizer(stop_words="english")

X_train_bow = vectorizer_bow.fit_transform(X_train)
X_test_bow = vectorizer_bow.transform(X_test)

# --- NAIVE BAYES MODEL ---
model_bow_nb = MultinomialNB()
model_bow_nb.fit(X_train_bow, y_train)

# --- PREDIKCIJE ---
y_pred_bow_nb = model_bow_nb.predict(X_test_bow)

# --- REZULTATI ---
print("Accuracy with BoW + MultinomialNB:", accuracy_score(y_test, y_pred_bow_nb))
print("\nClassification report with BoW + MultinomialNB:\n")
print(classification_report(y_test, y_pred_bow_nb))
```

```
Accuracy with BoW + MultinomialNB: 0.6666666666666666

Classification report with BoW + MultinomialNB:

              precision    recall  f1-score   support

    business       0.50      1.00      0.67         3
    politics       1.00      0.33      0.50         3
      sports       0.50      0.33      0.40         3
  technology       1.00      1.00      1.00         3

    accuracy                           0.67        12
   macro avg       0.75      0.67      0.64        12
weighted avg       0.75      0.67      0.64        12
```

```
# --- BAG OF WORDS + N-GRAMS ---
vectorizer_ng = CountVectorizer(stop_words="english", ngram_range=(1,2))

X_train_ng = vectorizer_ng.fit_transform(X_train)
X_test_ng = vectorizer_ng.transform(X_test)

# --- NAIVE BAYES MODEL ---
model_ng_nb = MultinomialNB()
model_ng_nb.fit(X_train_ng, y_train)

# --- PREDIKCIJE ---
y_pred_ng_nb = model_ng_nb.predict(X_test_ng)

# --- REZULTATI ---
print("Accuracy with n-grams + MultinomialNB:", accuracy_score(y_test, y_pred_ng_nb))
print("\nClassification report with n-grams + MultinomialNB:\n")
print(classification_report(y_test, y_pred_ng_nb))
```

```
Accuracy with n-grams + MultinomialNB: 0.6666666666666666

Classification report with n-grams + MultinomialNB:

              precision    recall  f1-score   support

    business       0.50      1.00      0.67         3
    politics       1.00      0.33      0.50         3
      sports       0.50      0.33      0.40         3
  technology       1.00      1.00      1.00         3

    accuracy                           0.67        12
   macro avg       0.75      0.67      0.64        12
weighted avg       0.75      0.67      0.64        12
```

```
new_articles = [
    "The minister discussed tax reform and foreign policy during a parliamentary debate.",
    "The striker scored twice and the team won the match in the final minutes.",
    "The company released a software patch to fix a serious security issue.",
    "The bank reported stronger earnings and lower operating costs this quarter."
]

new_X = vectorizer_ng.transform(new_articles)
predictions = model_ng.predict(new_X)
```